

Catching the Third moment in intraday level

Trexquant week 3 Notebook Strategy

May 22, 2024

Jack(Wensheng) Li

Motivation

- When considering the distributions of asset returns, investors are typically focused on the **first and second moments** of those distributions
- Skewness**, the third moment, is not often taken into consideration
- Different Stocks may have the same mean and std. But different skewness shows different future returns.

Realized moments and the cross-sectional stock returns around earnings announcements

Qingxia Wang,^a Robert Faff,^b Min Zhu,^c

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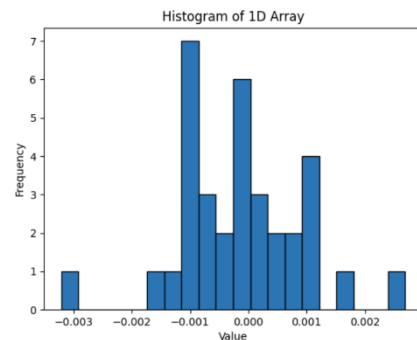
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Abstract

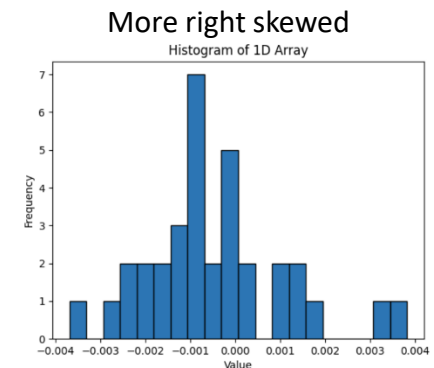
We examine the predictability of realized measures on the cross-section of stock returns around earnings announcements. We construct realized measures (variance, skewness, kurtosis, and relative jump) using high-frequency intraday stock prices. Our results show that realized variance, skewness, and relative jumps strongly predict stock returns around earnings announcements but realized kurtosis does not. These findings are robust to various event windows, after controlling for firm characteristics, and to a range of

Intra_cps5_x distribution of 2 stocks on the 3550th day



```
[72]: nn.describe()
```

	6	0
count	34.000000	34.000000
mean	-0.000443	0.000169
std	0.001594	0.001286



Shows higher future return

Some trials

- Incorporate skewness of intraday log returns as features— —didn't significantly improve
- ts_zscore and ts_mean do not help
- Although it performs well as a standalone alpha in the backtesting system.

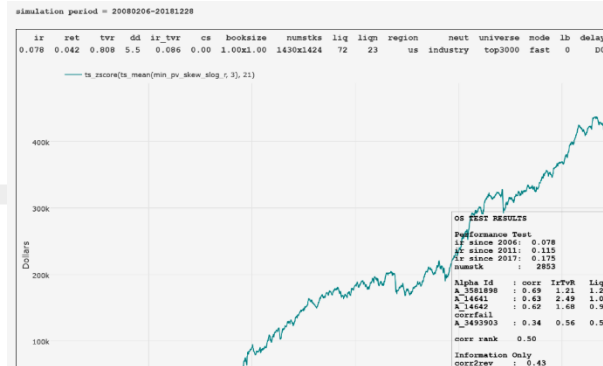
	accuracy	ir	ret	tvr	dd	liqn
train	51.408	0.074	0.120	0.368	16.369	43.186
valid	51.288	0.077	0.112	0.328	7.524	43.186


```

1
2 alpha = ts_zscore(ts_mean(min_pv_skew_slog_r, 3), 21)
3 # alpha =
  
```



- Many research reports have pointed out that
- Before performance announcements, uninformed traders show risk aversion towards assets with **high absolute skewness, regardless of its direction.**
- Informed traders trade **more aggressively** before announcements, **influencing trading volume directly.**
- **The velocity of information flow** among informed traders correlates strongly with future stock returns.



Momentum and post-earnings-announcement drift anomalies: The role of liquidity risk[☆]

Ronnie Sadka*

Department of Finance and Business Economics, University of Washington Business School, Seattle, WA 98195-3200, USA

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Abstract

This paper investigates the components of liquidity risk that are important for understanding asset-pricing anomalies. Firm-level liquidity is decomposed into variable and fixed price effects and estimated using intraday data for the period 1983–2001. Unexpected systematic (market-wide) variations of the variable component rather than the fixed component of liquidity are shown to be priced within the context of momentum and post-earnings-announcement drift (PEAD) portfolio returns. As the variable component is typically associated with private information [e.g., Kyle, 1985, *Econometrica* 53, 1315–1335], the results suggest that a substantial part of momentum and PEAD returns can be viewed as compensation for the unexpected variations in the aggregate ratio of informed traders to noise traders.
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Catching the (Third) Moment

When considering the distributions of asset returns, investors are typically focused on the first and second moments of those distributions: the average returns and their standard deviation are the characteristics center of attention. Skewness, the third moment, is not often taken into consideration a priori. However, if outsized returns from a particular asset hit the portfolio—either positive or negative—this asset is often shunned in case of negative returns or excessively appreciated when that outsized return is positive. This overreaction may well have its roots in investors evaluating risk according to the Prospect Theory (Tversky and Kahneman, 1992). As a consequence investors

Two main findings

- pure volume, volume adjusted return > intraday return only (for skewness)
- STD > mean (on daily level) } Only before announcement

min_pv_skew_slog_r_mean								'macd_30min_180min_skew_volume_1d_mean'								macd_30min_180min_skew_volume_1d_std						
	accuracy	ir	ret	tvr	dd	liqn			accuracy	ir	ret	tvr	dd	liqn			accuracy	ir	ret	tvr	dd	liqn
train	51.408	0.074	0.120	0.368	16.369	43.186	⇒	train	51.472	0.087	0.139	0.365	13.887	42.786	⇒	train	51.460	0.096	0.145	0.360	13.119	44.673
valid	51.288	0.077	0.112	0.328	7.524	43.186		valid	51.380	0.089	0.131	0.326	6.901	42.786		valid	51.676	0.094	0.140	0.325	6.224	44.673

↓ better

'min_pv_skew_sqrt_abs_r_mean'								downvolume_retskew_allday_mean'								downvolume_retskew_allday_std						
	accuracy	ir	ret	tvr	dd	liqn			accuracy	ir	ret	tvr	dd	liqn			accuracy	ir	ret	tvr	dd	liqn
train	51.394	0.091	0.146	0.365	13.944	42.422	⇒	train	51.467	0.089	0.141	0.367	17.566	42.586	⇒	train	51.551	0.107	0.163	0.362	12.665	43.939
valid	51.434	0.084	0.128	0.328	8.339	42.422		valid	51.192	0.091	0.130	0.328	7.302	42.586		valid	51.362	0.091	0.131	0.324	5.836	43.939

⇒

skew_volume_afternoon_mean								skew_volume_afternoon_std						
	accuracy	ir	ret	tvr	dd	liqn			accuracy	ir	ret	tvr	dd	liqn
train	51.385	0.090	0.143	0.364	14.309	42.152	⇒							
valid	51.394	0.090	0.130	0.330	6.280	42.152								

⇒

num_tc_buy_emo_tsz_10min_skw_mean								num_tc_buy_emo_tsz_10min_skw_std						
	accuracy	ir	ret	tvr	dd	liqn			accuracy	ir	ret	tvr	dd	liqn
train	51.353	0.084	0.130	0.363	13.340	41.887								
valid	51.514	0.092	0.131	0.326	5.662	41.887								

Better but highly correlated with above two

Better but highly correlated with above two

Some Reasons

- Skewness, measuring extreme return probabilities, the sign becomes less important to investors on the eve of announcements.
- Informed traders react to insider information and they **must** to do sth, increasing **absolute value of skewness or its std more**, resulting in a negative correlation with performance jumps.

skew_volume_afternoon_mean

	accuracy	ir	ret	tvr	dd	liqn
train	51.385	0.090	0.143	0.364	14.309	42.152
valid	51.394	0.090	0.130	0.330	6.280	42.152

num_tc_buy_emo_tsz_10min_skw_mean

	accuracy	ir	ret	tvr	dd	liqn
train	51.353	0.084	0.130	0.363	13.340	41.887
valid	51.514	0.092	0.131	0.326	5.662	41.887

downvolume_retskew_allday_mean'

	accuracy	ir	ret	tvr	dd	liqn
train	51.467	0.089	0.141	0.367	17.566	42.586
valid	51.192	0.091	0.130	0.328	7.302	42.586

macd_30min_180min_skew_volume_1d_std

	accuracy	ir	ret	tvr	dd	liqn
train	51.460	0.096	0.145	0.360	13.119	44.673
valid	51.676	0.094	0.140	0.325	6.224	44.673

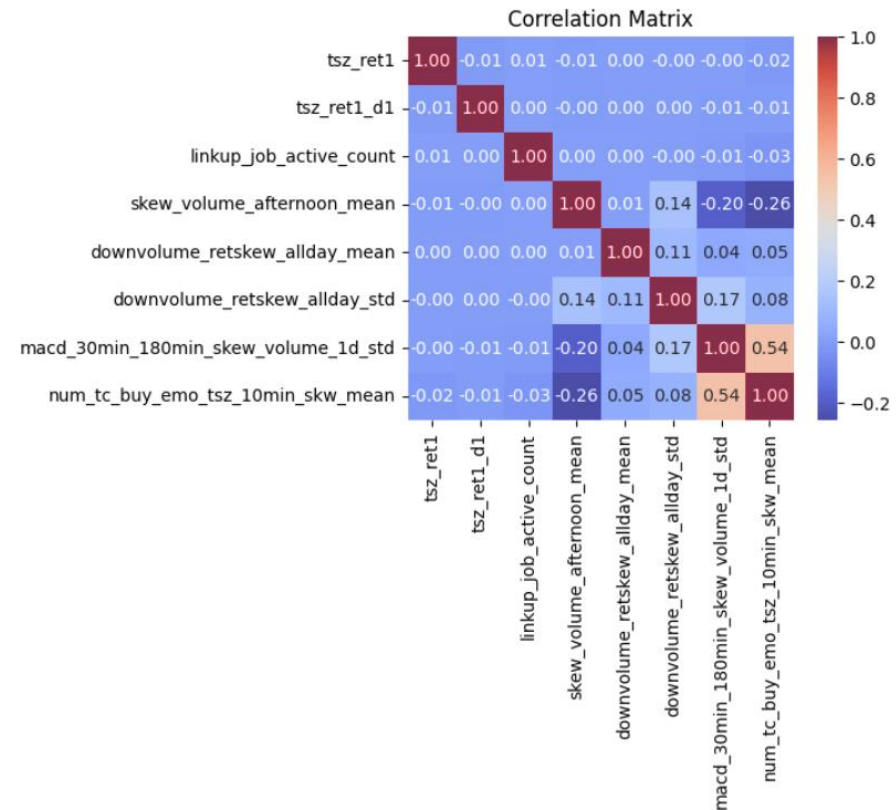
downvolume_retskew_allday_std

	accuracy	ir	ret	tvr	dd	liqn
train	51.551	0.107	0.163	0.362	12.665	43.939
valid	51.362	0.091	0.131	0.324	5.836	43.939

Results

- I ultimately chose to use the skewness of intraday volume-lighted/adjusted returns and the skewness of intraday volume as our features.
- I predominantly utilized ts_std (some use ts_mean) to process the aforementioned intraday indicators,
- Therefore, the final results are as follows:

	accuracy	ir	ret	tvr	dd	liqn
train	51.673	0.116	0.170	0.358	13.214	45.417
valid	51.564	0.112	0.159	0.324	5.669	45.417



Thanks

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Jack(Wensheng) Li